BCA201P — Foundations of Java Development

Lab Sheet 01 • Java Assignment Solutions

BCA III Semester | Session 2026–2027

30 Java programs covering variables, operators, decision-making, and basic loops.

GitHub-ready PDF reference

1. Display name, age, and college name

```
public class Main {
    public static void main(String[] args) {
        String name = "Your Name";
        int age = 20;
        String college = "Your College Name";

        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("College: " + college);
    }
}
```

2. Arithmetic operations on two integers

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two integers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        if (b != 0) {
            System.out.println("Division: " + ((double) a / b));
            System.out.println("Modulus: " + (a % b));
        } else {
            System.out.println("Division and modulus by zero are undefined.");
        }
        sc.close();
    }
}
```

3. Check even or odd

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (n % 2 == 0)
            System.out.println("Even number");
        else
            System.out.println("Odd number");

        sc.close();
    }
}
```

4. Demonstrate relational operators

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two numbers: ");
        int a = sc.nextInt(), b = sc.nextInt();

        System.out.println("a > b: " + (a > b));
        System.out.println("a < b: " + (a < b));
        System.out.println("a >= b: " + (a >= b));
        System.out.println("a <= b: " + (a <= b));
        System.out.println("a == b: " + (a == b));
        System.out.println("a != b: " + (a != b));
        sc.close();
    }
}
```

5. Explicitly convert float to integer

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a floating-point number: ");
        double value = sc.nextDouble();
        int converted = (int) value;

        System.out.println("Original value: " + value);
        System.out.println("Converted integer: " + converted);
        sc.close();
    }
}
```

6. Print ASCII value of a character

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        System.out.println("Character: " + ch);
        System.out.println("Numeric code: " + (int) ch);
        sc.close();
    }
}
```

7. Check divisibility by both 3 and 5

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (n % 3 == 0 && n % 5 == 0)
            System.out.println("Divisible by both 3 and 5");
        else
            System.out.println("Not divisible by both 3 and 5");

        sc.close();
    }
}
```

8. Check vowel or consonant

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an alphabet: ");
        char ch = Character.toLowerCase(sc.next().charAt(0));

        if (ch < 'a' || ch > 'z')
            System.out.println("Not an alphabet");
        else if ("aeiou".indexOf(ch) >= 0)
            System.out.println("Vowel");
        else
            System.out.println("Consonant");

        sc.close();
    }
}
```

9. Total, percentage, and pass/fail for three subjects

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter marks in three subjects (out of 100): ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();

        int total = a + b + c;
        double percentage = total / 3.0;
        System.out.println("Total: " + total);
        System.out.println("Percentage: " + percentage + "%");

        if (a >= 40 && b >= 40 && c >= 40)
            System.out.println("Pass");
        else
            System.out.println("Fail");

        sc.close();
    }
}
```

10. Greatest of two numbers

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two numbers: ");
        int a = sc.nextInt(), b = sc.nextInt();

        if (a > b)
            System.out.println(a + " is greater");
        else if (b > a)
            System.out.println(b + " is greater");
        else
            System.out.println("Both numbers are equal");

        sc.close();
    }
}
```

11. Largest of three numbers using nested if-else

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three numbers: ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();

        if (a >= b) {
            if (a >= c)
                System.out.println("Largest: " + a);
            else
                System.out.println("Largest: " + c);
        } else {
            if (b >= c)
                System.out.println("Largest: " + b);
            else
                System.out.println("Largest: " + c);
        }
        sc.close();
    }
}
```

12. Check leap year

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a year: ");
        int year = sc.nextInt();

        if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
            System.out.println("Leap year");
        else
            System.out.println("Not a leap year");

        sc.close();
    }
}
```

13. Check positive, negative, or zero

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (n > 0)
            System.out.println("Positive");
        else if (n < 0)
            System.out.println("Negative");
        else
            System.out.println("Zero");

        sc.close();
    }
}
```

14. Classify a character

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if (Character.isDigit(ch))
            System.out.println("Digit");
        else if (Character.isUpperCase(ch))
            System.out.println("Uppercase letter");
        else if (Character.isLowerCase(ch))
            System.out.println("Lowercase letter");
        else
            System.out.println("Special character");

        sc.close();
    }
}
```

15. Check a three-digit Armstrong number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a three-digit number: ");
        int n = sc.nextInt();
        int original = n, sum = 0;

        while (n > 0) {
            int digit = n % 10;
            sum += digit * digit * digit;
            n /= 10;
        }

        if (original >= 100 && original <= 999 && sum == original)
            System.out.println("Armstrong number");
        else
            System.out.println("Not an Armstrong number");

        sc.close();
    }
}
```

16. Calculate tax using the given income slabs

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter income: ");
        double income = sc.nextDouble();
        double tax;

        if (income <= 250000)
            tax = 0;
        else if (income <= 500000)
            tax = income * 0.05;
        else if (income <= 1000000)
            tax = income * 0.20;
        else
            tax = income * 0.30;

        System.out.println("Tax: " + tax);
        sc.close();
    }
}
```

17. Check integer palindrome

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n = sc.nextInt();
        int original = n, reversed = 0;

        while (n > 0) {
            reversed = reversed * 10 + n % 10;
            n /= 10;
        }

        if (original >= 0 && original == reversed)
            System.out.println("Palindrome");
        else
            System.out.println("Not a palindrome");

        sc.close();
    }
}
```

18. Check prime number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        boolean prime = n >= 2;

        for (int i = 2; i <= n / i && prime; i++) {
            if (n % i == 0)
                prime = false;
        }

        System.out.println(prime ? "Prime number" : "Not a prime number");
        sc.close();
    }
}
```

19. Perform an operation selected by an operator character

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two numbers: ");
        double a = sc.nextDouble(), b = sc.nextDouble();
        System.out.print("Enter operator (+, -, *, /): ");
        char op = sc.next().charAt(0);

        if (op == '+')
            System.out.println("Result: " + (a + b));
        else if (op == '-')
            System.out.println("Result: " + (a - b));
        else if (op == '*')
            System.out.println("Result: " + (a * b));
        else if (op == '/' && b != 0)
            System.out.println("Result: " + (a / b));
        else
            System.out.println("Invalid operator or division by zero");

        sc.close();
    }
}
```

20. Marriage eligibility by age and gender

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter age: ");
        int age = sc.nextInt();
        System.out.print("Enter gender (male/female): ");
        String gender = sc.next().toLowerCase();

        if (gender.equals("male")) {
            if (age >= 21)
                System.out.println("Eligible by the assignment criteria");
            else
                System.out.println("Not eligible by the assignment criteria");
        } else if (gender.equals("female")) {
            if (age >= 18)
                System.out.println("Eligible by the assignment criteria");
            else
                System.out.println("Not eligible by the assignment criteria");
        } else {
            System.out.println("Invalid gender input");
        }
        sc.close();
    }
}
```

21. Check whether three angles form a valid triangle

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three angles: ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();

        if (a > 0 && b > 0 && c > 0 && a + b + c == 180)
            System.out.println("Valid triangle");
        else
            System.out.println("Invalid triangle");

        sc.close();
    }
}
```

22. Check whether a rectangle is a square

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter length and breadth: ");
        double length = sc.nextDouble(), breadth = sc.nextDouble();

        if (length > 0 && length == breadth)
            System.out.println("It is a square");
        else
            System.out.println("It is not a square");

        sc.close();
    }
}
```

23. Convert Celsius to Fahrenheit

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();
        double fahrenheit = (double) (celsius * 9 / 5) + 32;

        System.out.println("Fahrenheit: " + fahrenheit);
        sc.close();
    }
}
```

24. Check perfect number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a positive integer: ");
        int n = sc.nextInt();
        int sum = 0;

        for (int i = 1; i <= n / 2; i++) {
            if (n % i == 0)
                sum += i;
        }

        if (n > 0 && sum == n)
            System.out.println("Perfect number");
        else
            System.out.println("Not a perfect number");

        sc.close();
    }
}
```

25. Apply a logical operator to two integers

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two integers: ");
        int a = sc.nextInt(), b = sc.nextInt();
        System.out.print("Enter logical operator (&, |, ^): ");
        char op = sc.next().charAt(0);

        if (op == '&')
            System.out.println("Result: " + (a != 0 && b != 0));
        else if (op == '|')
            System.out.println("Result: " + (a != 0 || b != 0));
        else if (op == '^')
            System.out.println("Result: " + ((a != 0) ^ (b != 0)));
        else
            System.out.println("Invalid logical operator");

        sc.close();
    }
}
```

26. Compare first and last digits of a five-digit number

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a five-digit number: ");
        int n = sc.nextInt();

        if (n >= 10000 && n <= 99999) {
            int first = n / 10000;
            int last = n % 10;
            if (first == last)
                System.out.println("First and last digits are the same");
            else
                System.out.println("First and last digits are different");
        } else {
            System.out.println("Please enter a five-digit positive number");
        }
        sc.close();
    }
}
```

27. Determine profit or loss

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter cost price and selling price: ");
        double cp = sc.nextDouble(), sp = sc.nextDouble();

        if (sp > cp)
            System.out.println("Profit: " + (sp - cp));
        else if (cp > sp)
            System.out.println("Loss: " + (cp - sp));
        else
            System.out.println("No profit, no loss");

        sc.close();
    }
}
```

28. Count the digits in an integer

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int n = sc.nextInt();
        long value = Math.abs((long) n);
        int count = 0;

        if (value == 0)
            count = 1;
        else {
            while (value > 0) {
                count++;
                value /= 10;
            }
        }

        System.out.println("Number of digits: " + count);
        sc.close();
    }
}
```

29. Classify height

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter height in cm: ");
        double height = sc.nextDouble();

        if (height < 150)
            System.out.println("Short");
        else if (height <= 170)
            System.out.println("Average");
        else
            System.out.println("Tall");

        sc.close();
    }
}
```

30. Simple calculator using if-else

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two numbers: ");
        double a = sc.nextDouble(), b = sc.nextDouble();
        System.out.print("Enter operator (+, -, *, /): ");
        char op = sc.next().charAt(0);

        if (op == '+')
            System.out.println("Result: " + (a + b));
        else if (op == '-')
            System.out.println("Result: " + (a - b));
        else if (op == '*')
            System.out.println("Result: " + (a * b));
        else if (op == '/') {
            if (b != 0)
                System.out.println("Result: " + (a / b));
            else
                System.out.println("Cannot divide by zero");
        } else {
            System.out.println("Invalid operator");
        }

        sc.close();
    }
}
```